

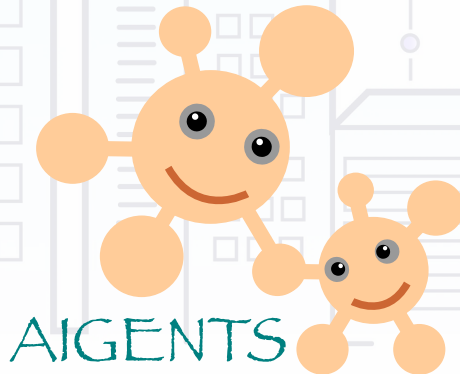
Aigents[®] for Cities and Communities

Anton Kolonin

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Telegram: [akolonin](#)

N* Novosibirsk
State
University
*THE REAL SCIENCE
<https://www.nsu.ru>

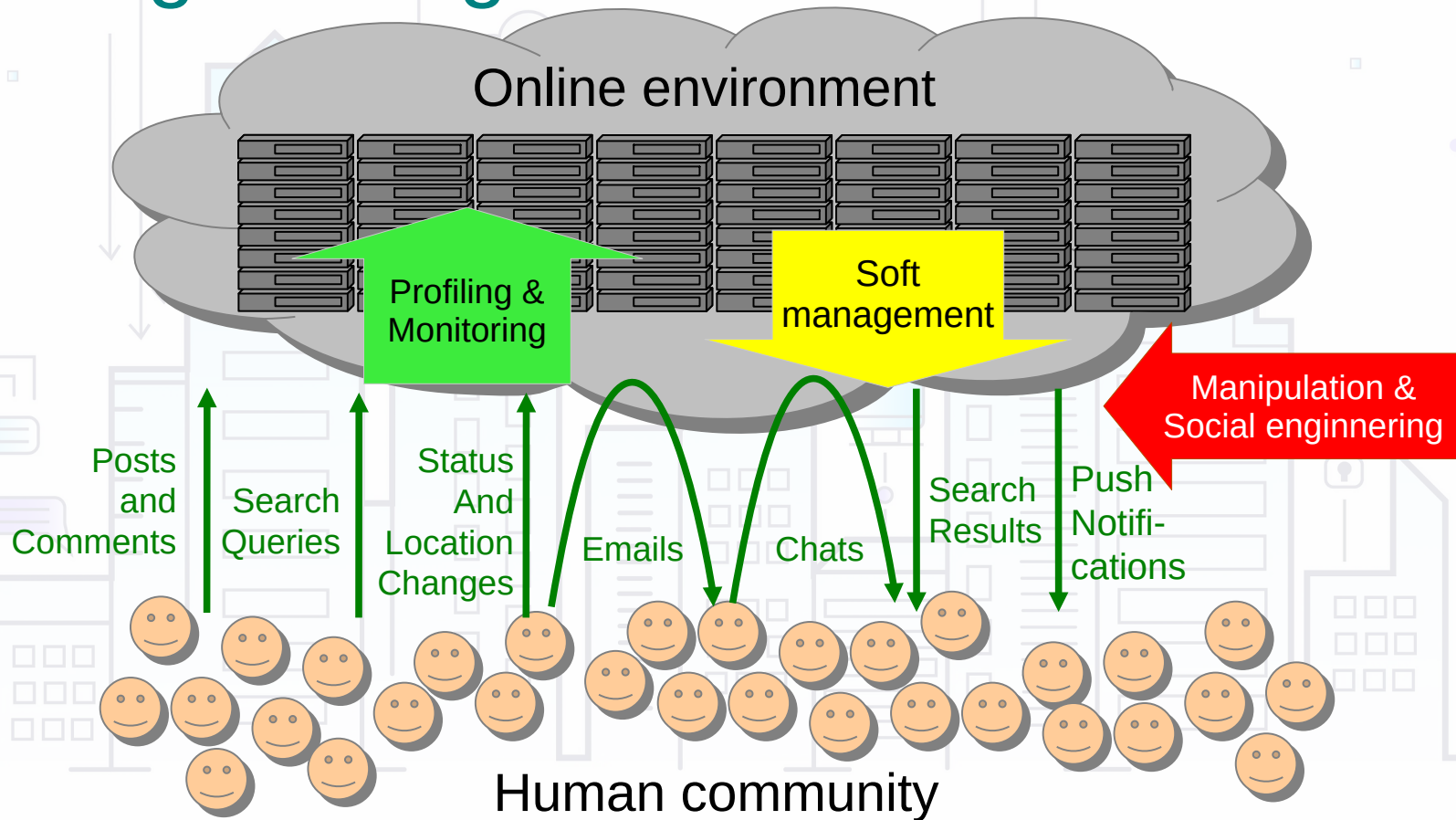


<https://aigents.com>



SingularityNET
<https://singularitynet.io>

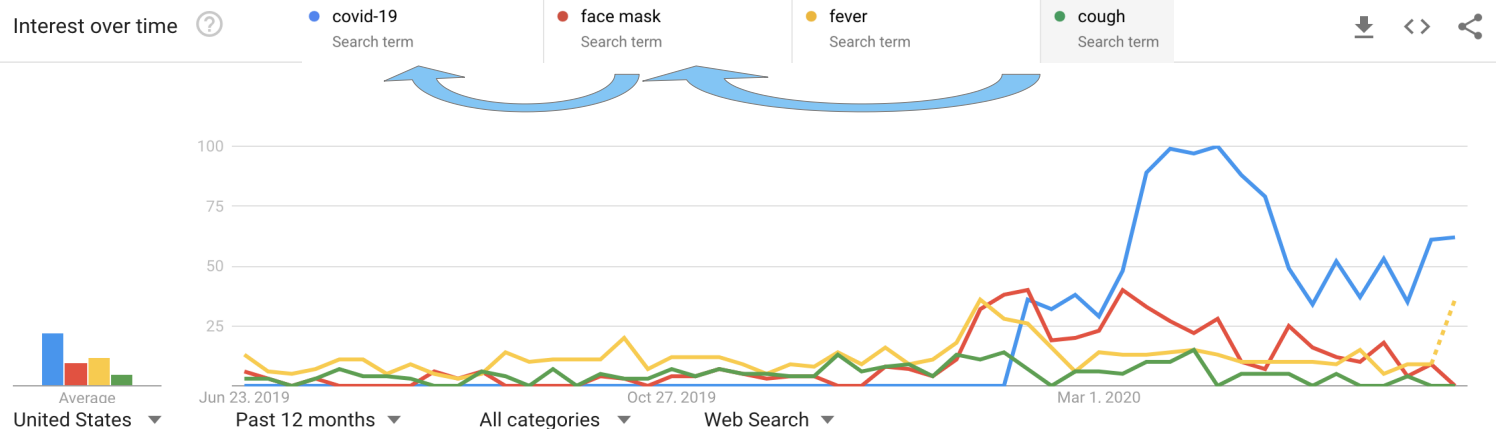
Architecture of collective intelligence and social engineering in online environments



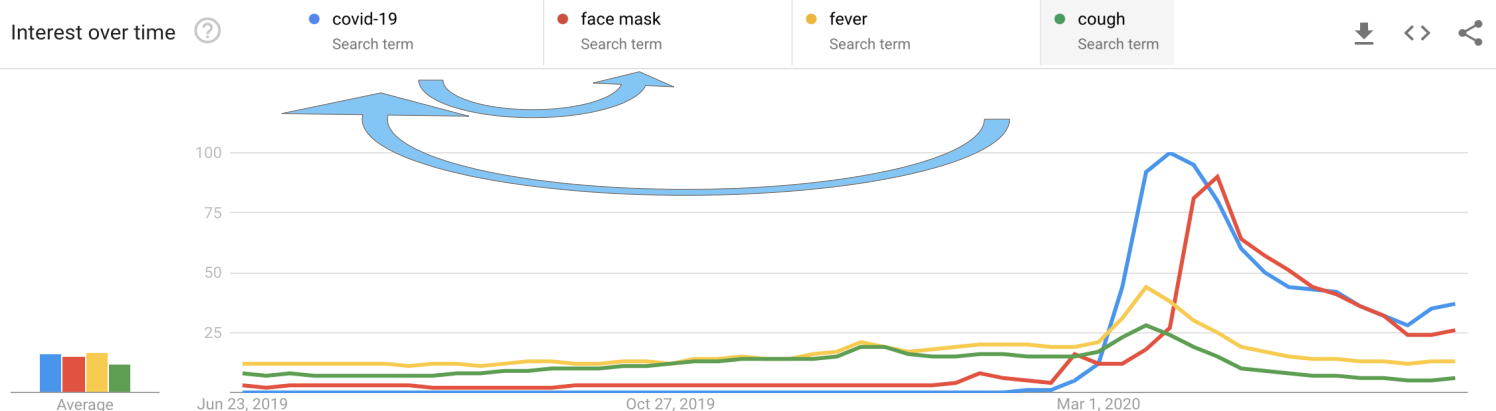
What do search queries talk about?

China ▼ Past 12 months ▼ All categories ▼ Web Search ▼

<https://trends.google.com/trends/explore?geo=CN&q=covid-19,face%20mask,fever,cough>

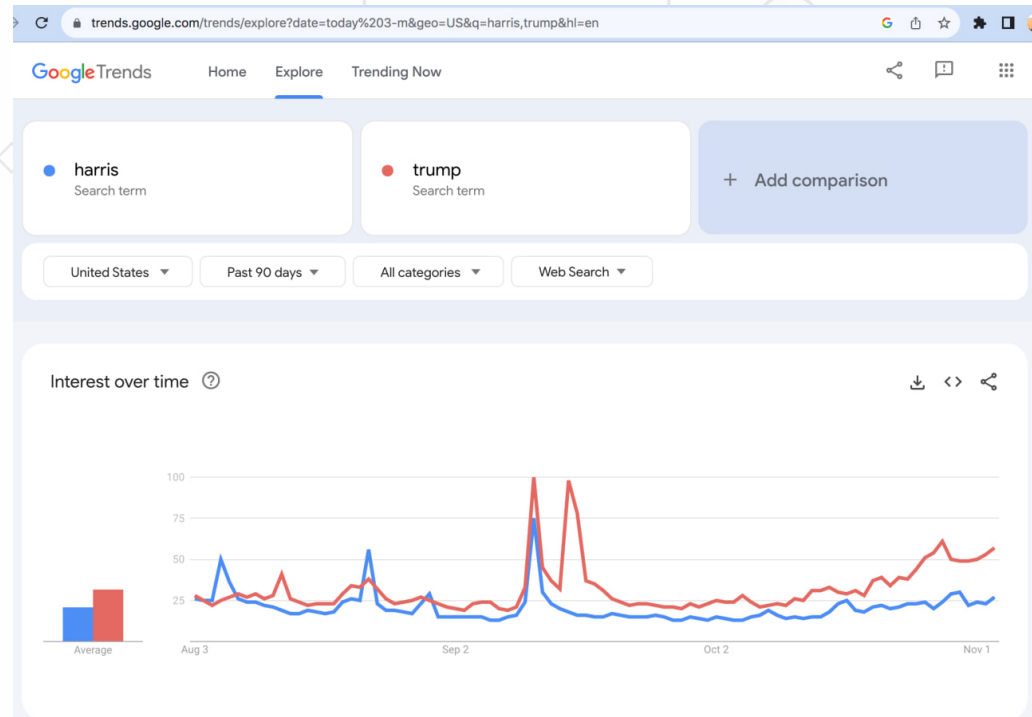


<https://trends.google.com/trends/explore?geo=US&q=covid-19,face%20mask,fever,cough>

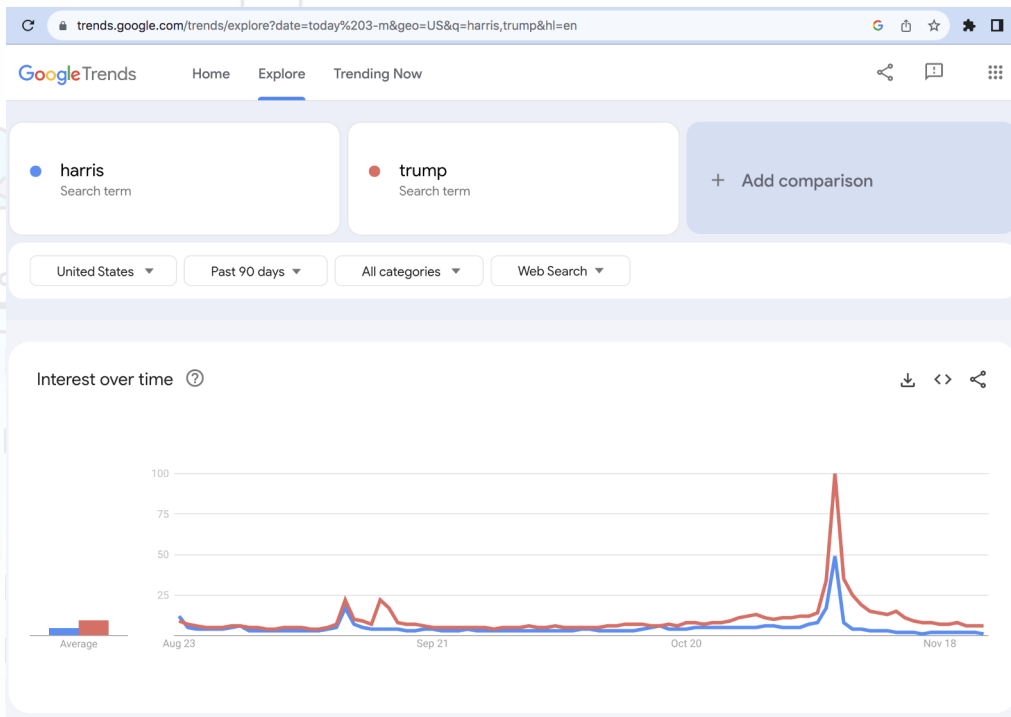


What do search queries talk about?

<https://trends.google.com/trends/explore?date=today%203-m&geo=US&q=harris,trump&hl=en>



Before US elections on November 5, 2024



On the day of elections and after

What can be learned about the psyche of a society from its texts?

Cognitive Distortions (Cognitive-Behavioral Therapy)

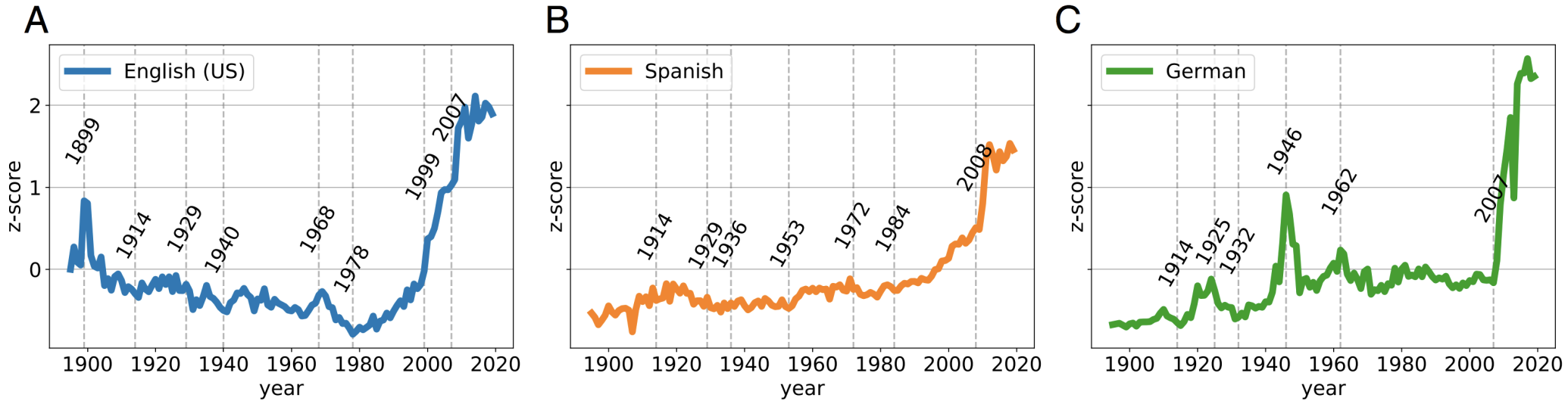
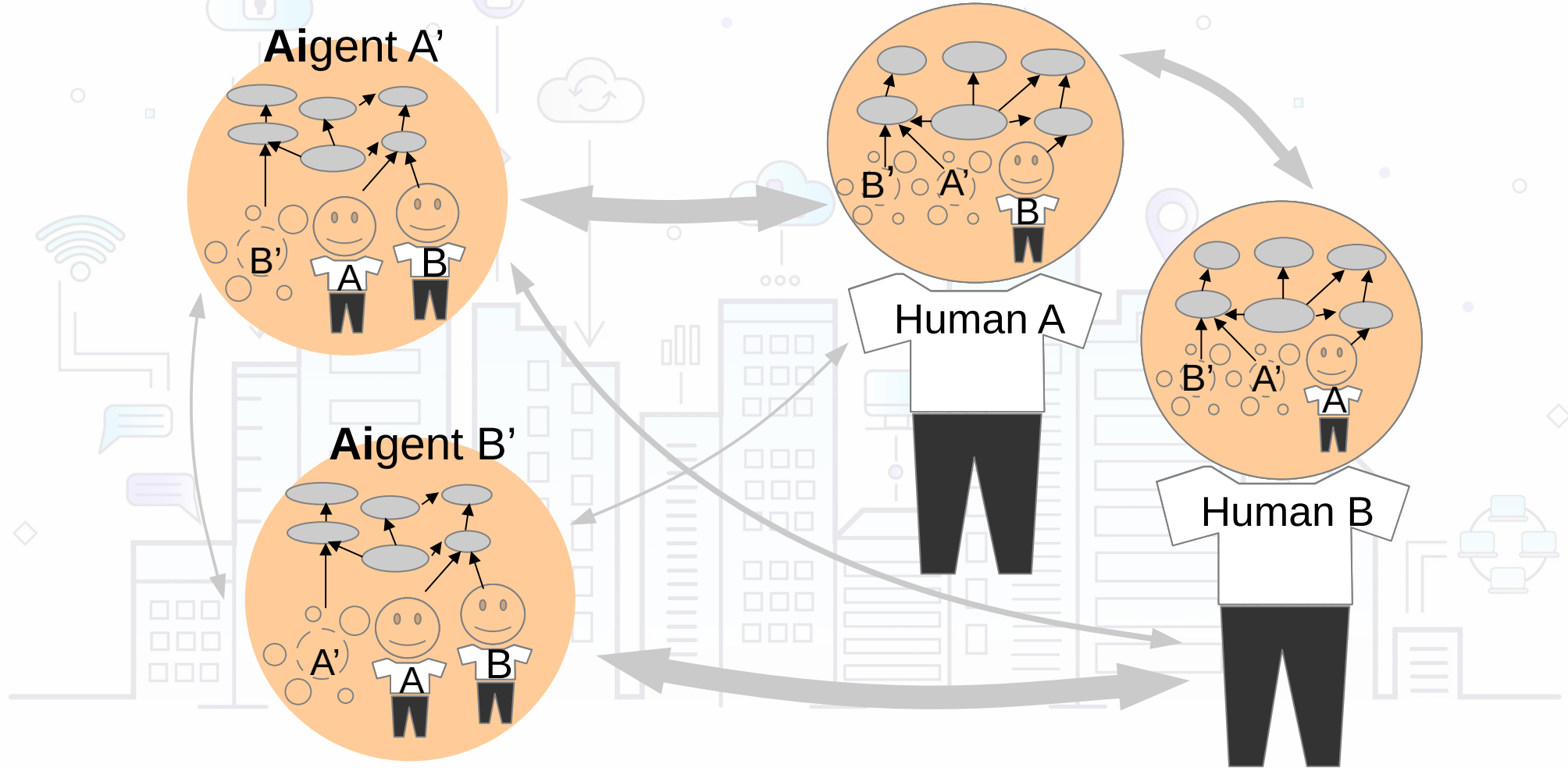


Fig. 2. (A–C) Median z scores of time series of CDS n-gram prevalence from 1855 to 2020 (125 y) in US English (A), Spanish (B), and German (C) with year markers added for major historical events. All time series reveal stable or declining levels for most of the 20th century followed by a sharp surge of cognitive distortions in the past three decades. US English shows declining levels from 1899 to 1978, with minor peaks around 1914 and 1940 (World War I and World War II) and notably 1968. This decline is followed by a surge of CDS prevalence starting in 1978 that continues to 2019. For Spanish we find stable levels from 1895 to the early 1980s at which point a trend occurs toward higher CDS prevalence levels above any of those previously observed. German shows stable CDS prevalence levels, with the exception of strong peaks around and after World War I and World War II, until 2007 at which point a sudden surge occurs.

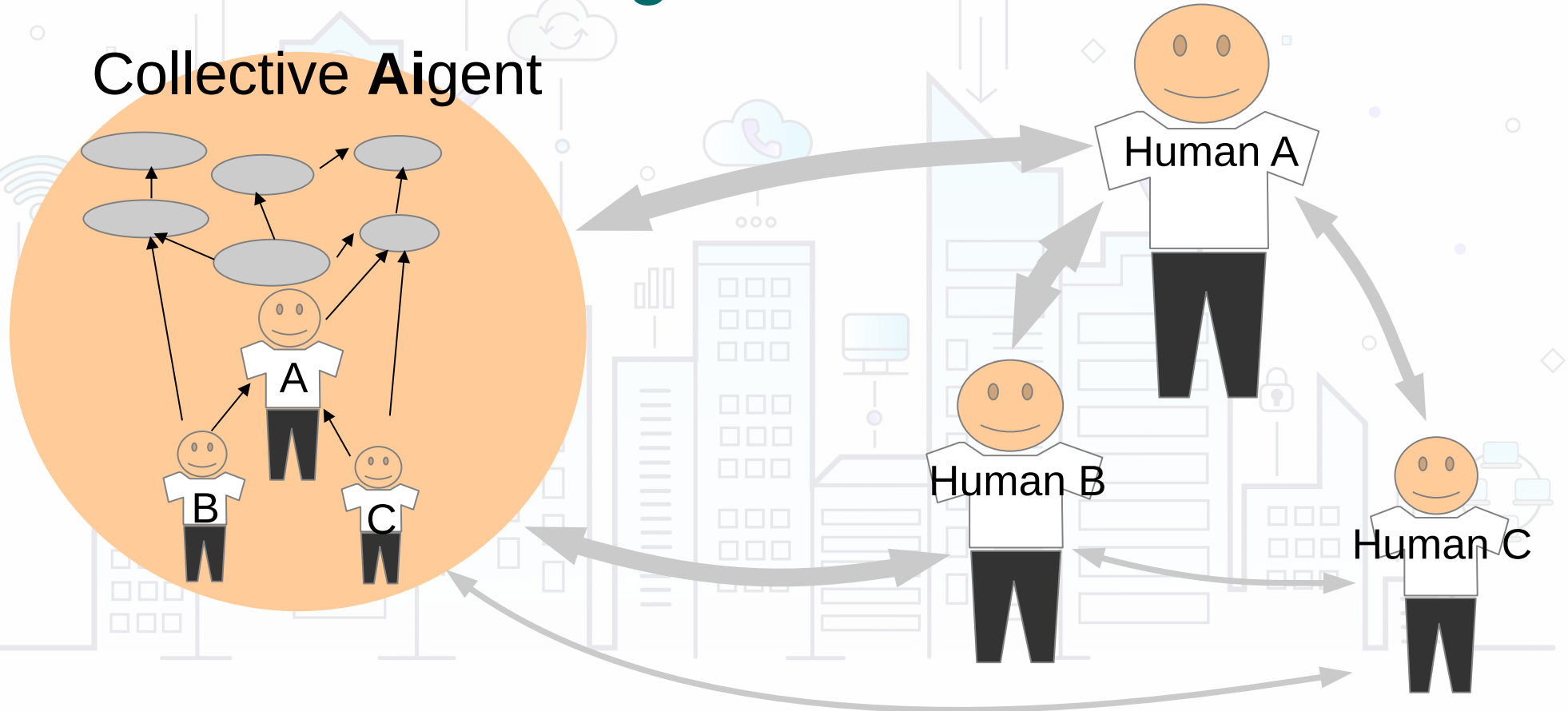
Source: <https://www.pnas.org/doi/pdf/10.1073/pnas.2102061118>

Socially-"aligned" emphatic AI-agents

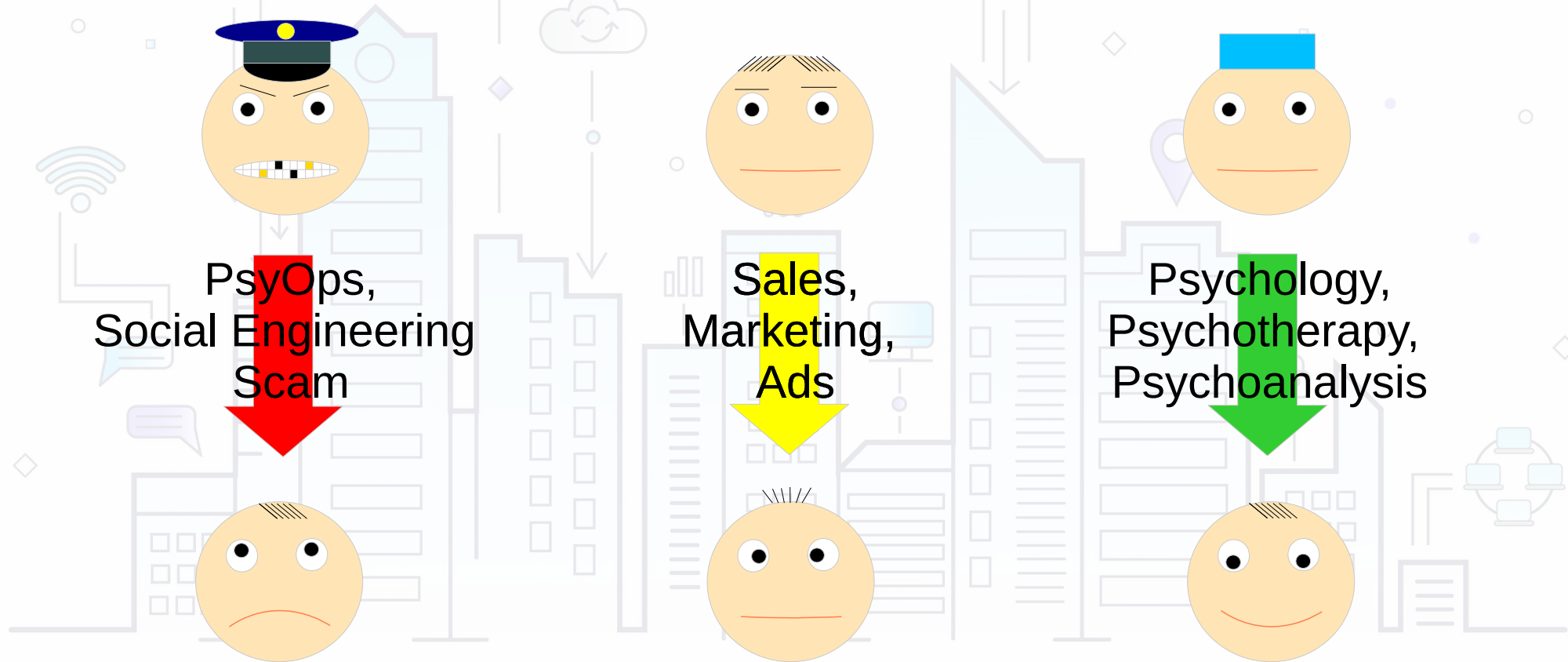


Decision support systems (DSS) relying on collective intelligence for communities

Collective Aigent

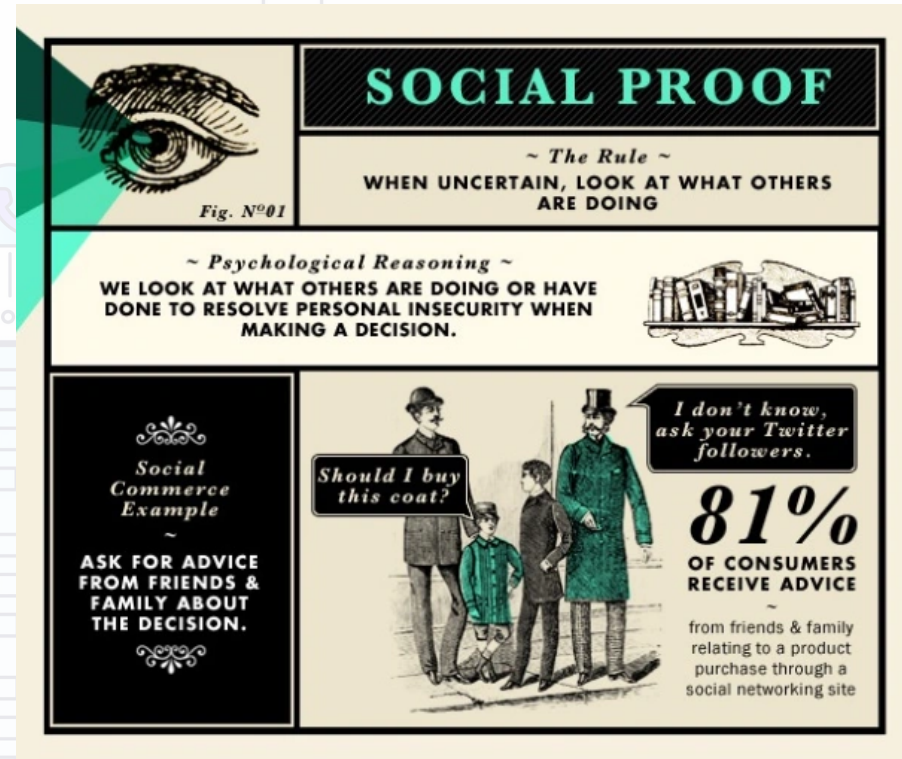


Enforcing/preventing manipulation (applied “belief engineering”)



Robert Cialdini: Social Proof

Social proof is a psychological and social phenomenon wherein people copy the actions of others in choosing how to behave in a given situation. The term was coined by Robert Cialdini in his 1984 book *Influence: Science and Practice*, and the concept is also known as informational social influence.



<https://www.amazon.com/Influence-Practice-Robert-B-Cialdini/dp/0205609996>

1984

Computable cognitive-behavioral model based on social evidence and restricted by resources

B.Goertzel, A.Kolonin, J.Pressing, C.Pennachin:
Compassion-based artificial psyche design
(1999-2000)

<https://www.goertzel.org/benzine/WakingUpFromTheEconomyOfDreams.htm>

A.Kolonin: Computable cognitive model based
on social evidence and restricted by resources
(2015)

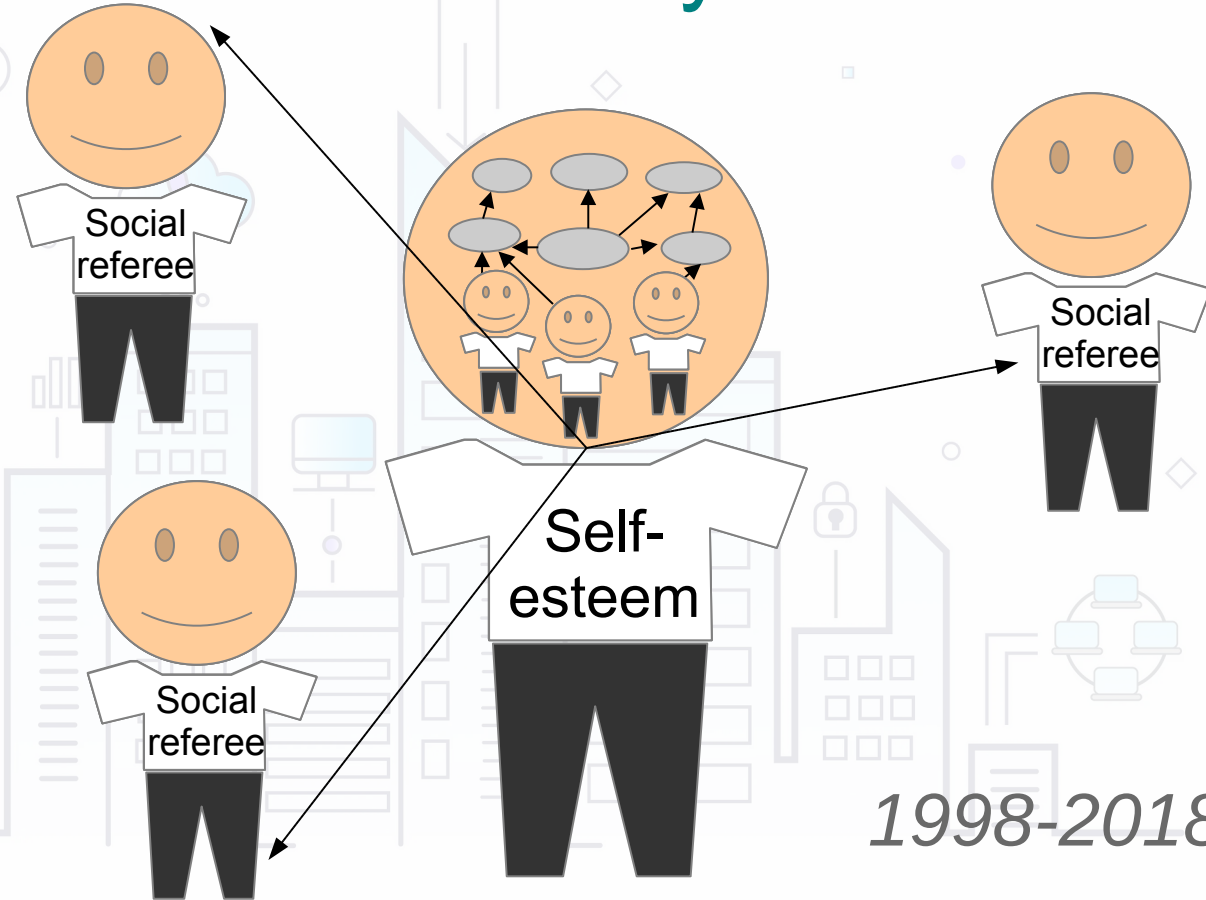
<https://ieeexplore.ieee.org/document/7361869>

Anton Kolonin, Evgenii Vityaev, Yuriy Orlov:
Cognitive Architecture of Collective Intelligence
Based on Social Evidence (2016)

<https://www.sciencedirect.com/science/article/pii/S1877050916317239>

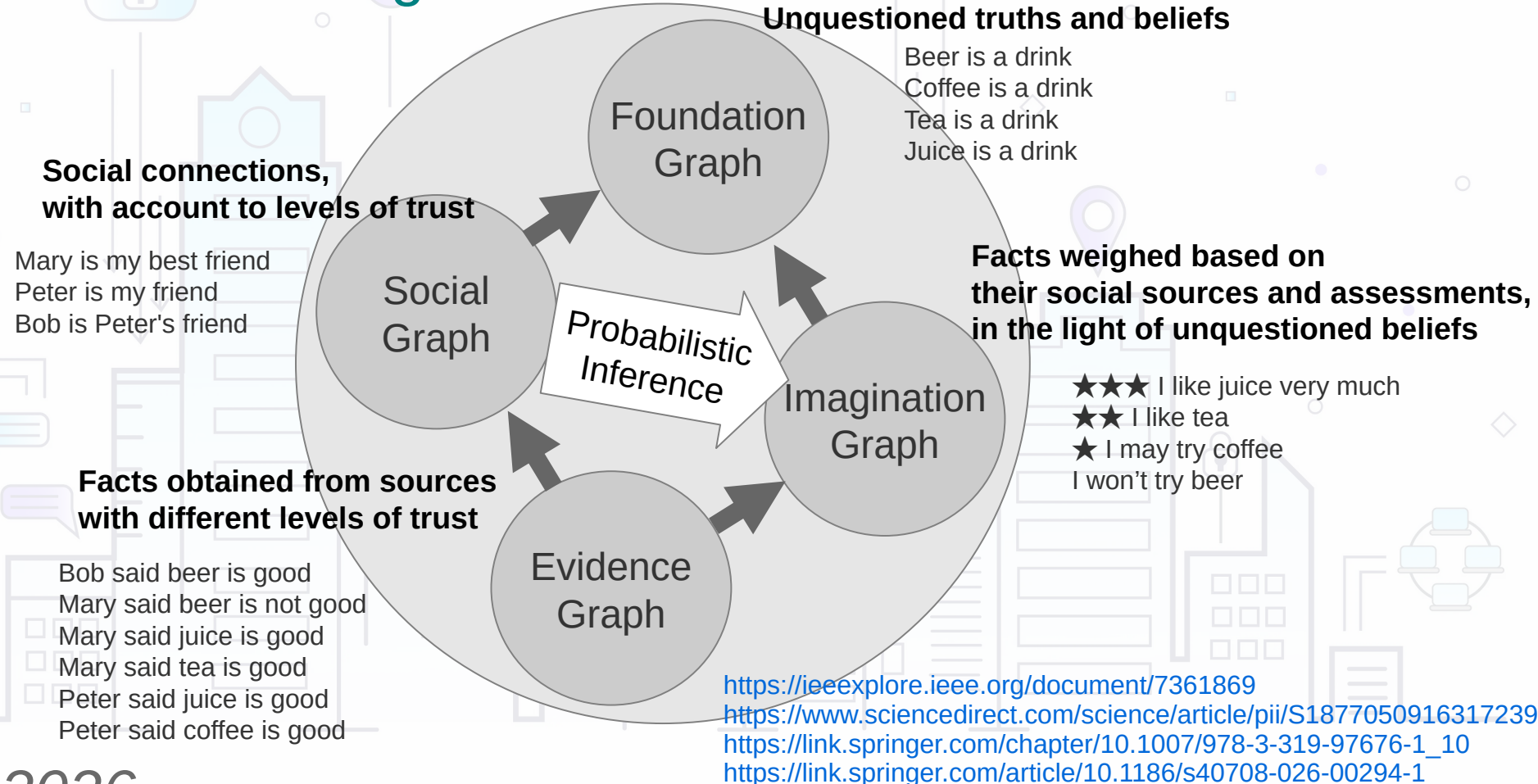
A.Kolonin: Resource-Constrained Social
Evidence Based Cognitive Model for Empathy-
Driven Artificial Intelligence (2018)

https://link.springer.com/chapter/10.1007/978-3-319-97676-1_10



1998-2018

• Social evidence-based resource-constrained cognitive-behavioral model



Perception & Action driven by Believed Social Evidence

i – person of consideration

j – concept (action) of consideration (exposed evidence)

l – person's belief item (of foundation graph of size L - personal preference base)

k – person's correspondent (of social graph of size K - social reference base)

B_{il} – person's i mental attachment to l (personal preference)

S_{ik} – person's i social bind to k (social reference)

$$P_{ij} = \sum_{l=1,L} (E^B_{ijl} * B_{il}) * \sum_{k=1,K} (E^S_{ijk} * S_{ik}) \quad (\text{what we think})$$

E^B_{ik} – concept (action) j agreement or compatibility with l in mind of i (believed evidence)

E^S_{ik} – concept (action) j expression or confirmation by k in view of i (social evidence)

P_{ij} – concept (action) j power for i (personal evidence)

$$C_{ij} = L + K + R^B + R^S \quad (\text{why we don't think about that})$$

C_{ij} – concept (action) cost j cost for i (personal cost)

• Social evidence-based resource-constrained cognitive-behavioral model

**Social connections,
with account to levels of trust**

S_{ik} – person's i social bind to k
 k – person's correspondent
 (of social graph of size K)

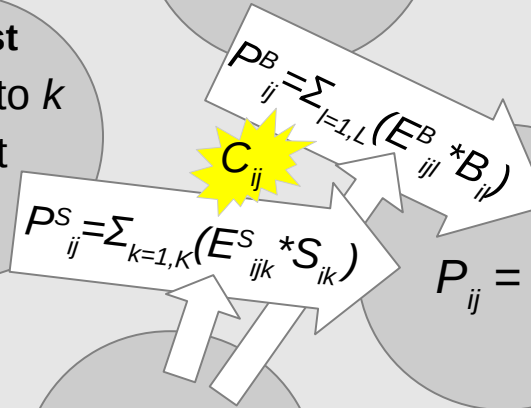
Unquestioned truths and beliefs

B_{il} – person's i mental attachment to l
 l – person's belief item
 (of foundation graph of size L)

**Facts weighed based on
their social sources and assessments,
in the light of unquestioned beliefs**

**Facts obtained from sources
with different levels of trust**

E_{ij} - exposed evidence



<https://ieeexplore.ieee.org/document/7361869>

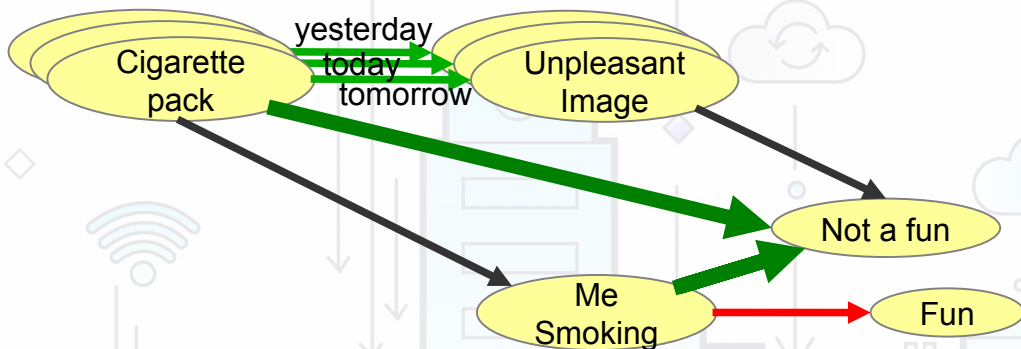
<https://www.sciencedirect.com/science/article/pii/S1877050916317239>

https://link.springer.com/chapter/10.1007/978-3-319-97676-1_10

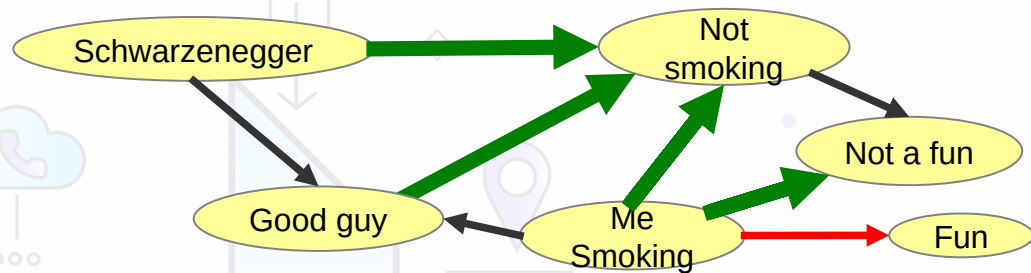
<https://link.springer.com/article/10.1186/s40708-026-00294-1>

Basic technique of applied “belief engineering”

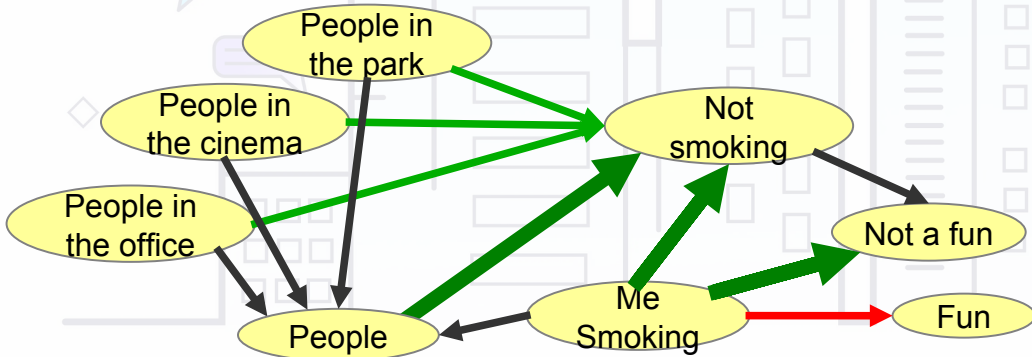
“Redundant temporal evidence”



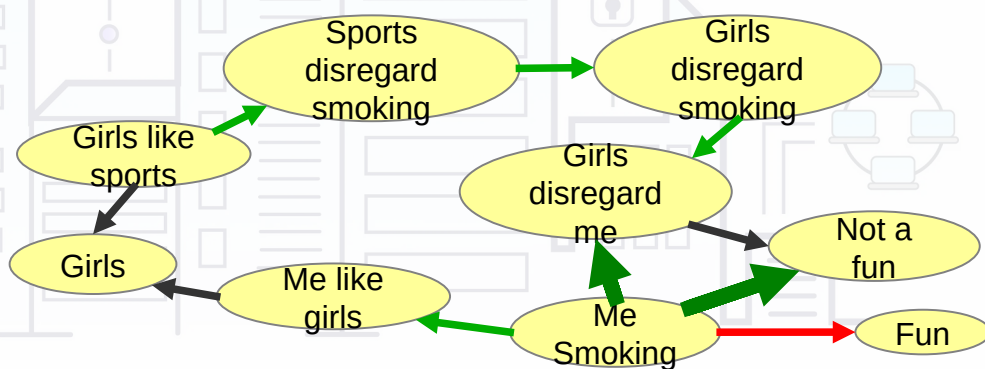
“Highly valuable social evidence”



“Redundant social evidence”

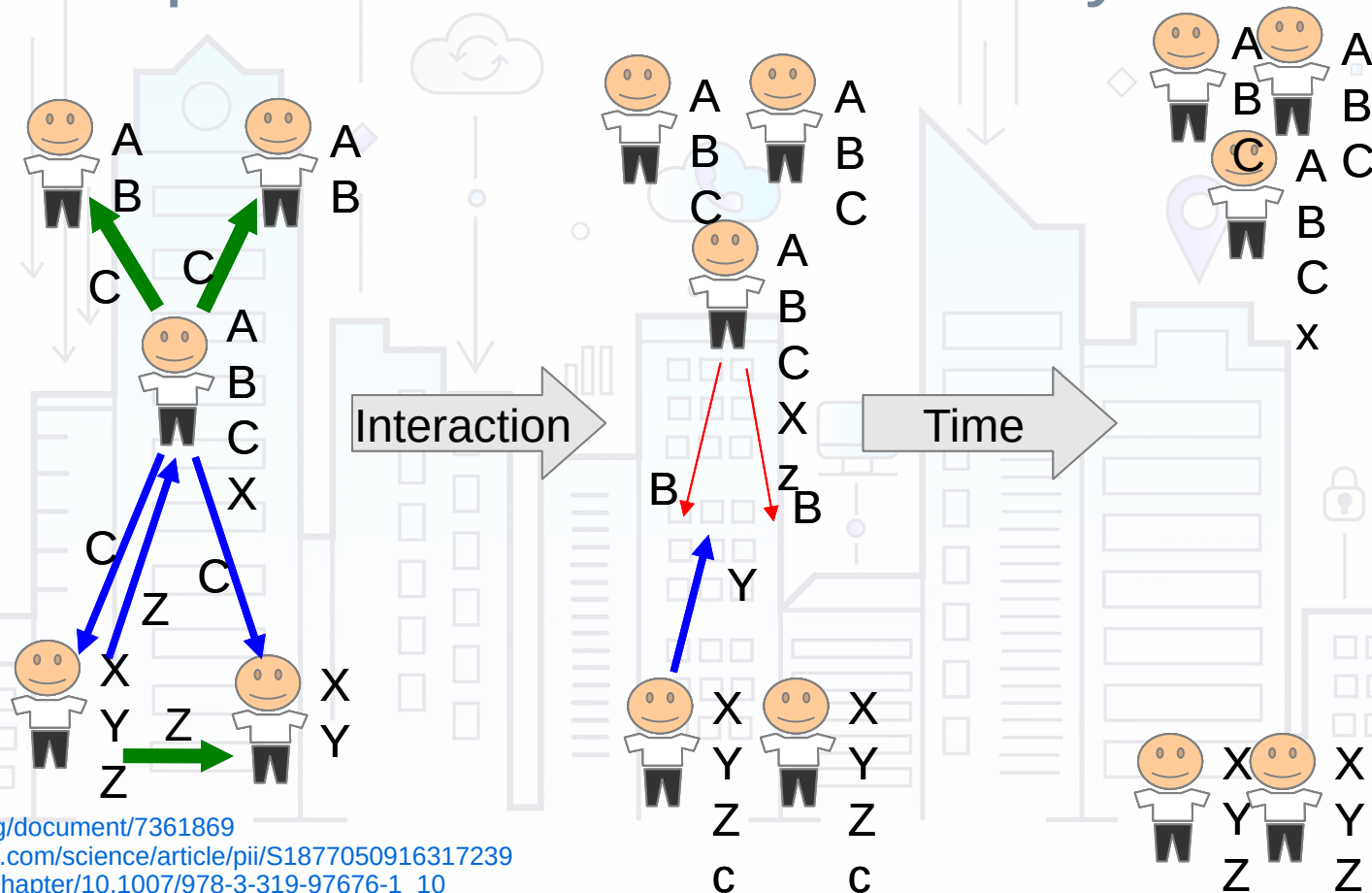


“Implicit evidence injection”



Social evidence-based cognitive-behavioral model

Helps to understand social dynamics



<https://ieeexplore.ieee.org/document/7361869>

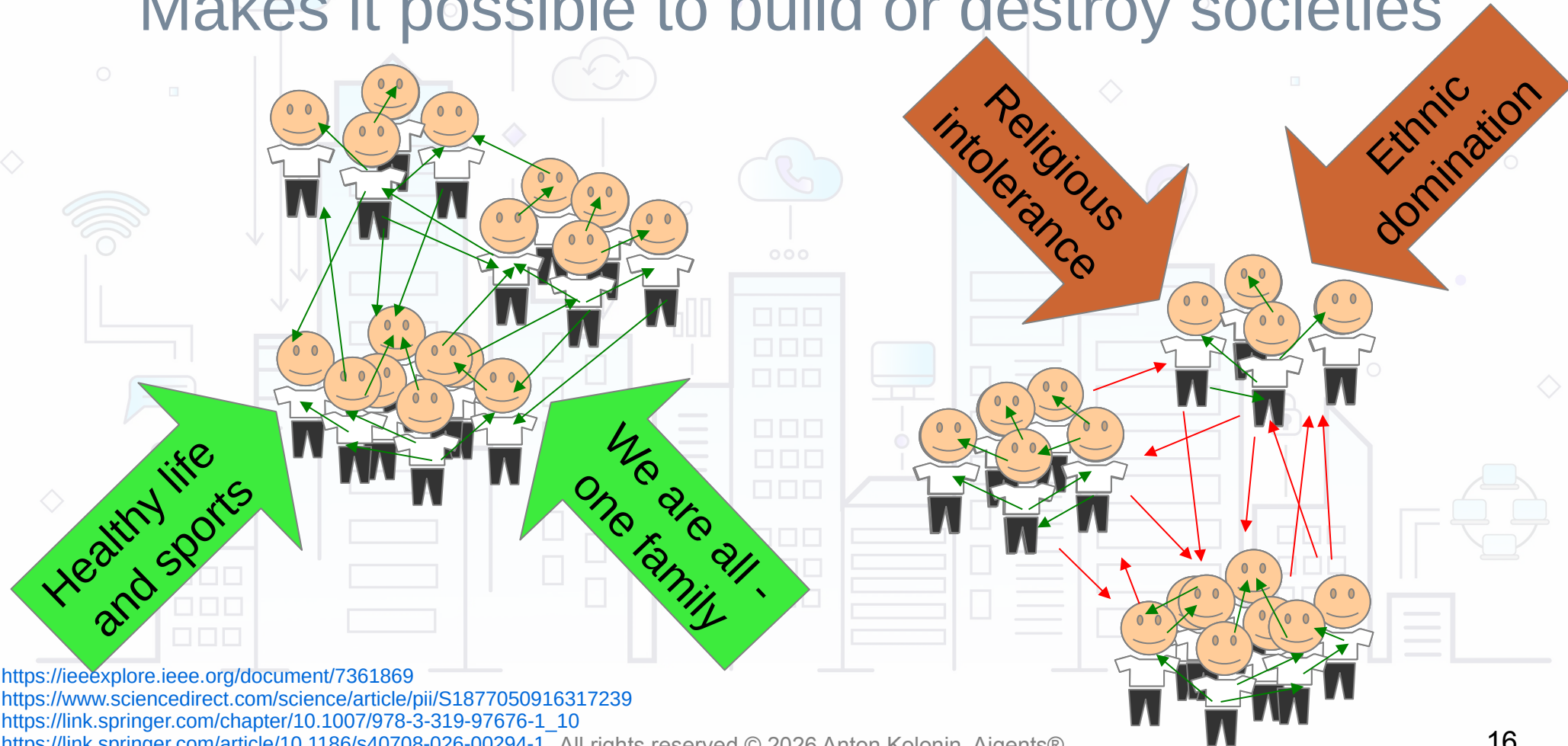
<https://www.sciencedirect.com/science/article/pii/S1877050916317239>

https://link.springer.com/chapter/10.1007/978-3-319-97676-1_10

<https://link.springer.com/article/10.1186/s40708-026-00294-1> All rights reserved © 2026 Anton Kolonin, Aigents®

Social evidence-based cognitive-behavioral model

Makes it possible to build or destroy societies



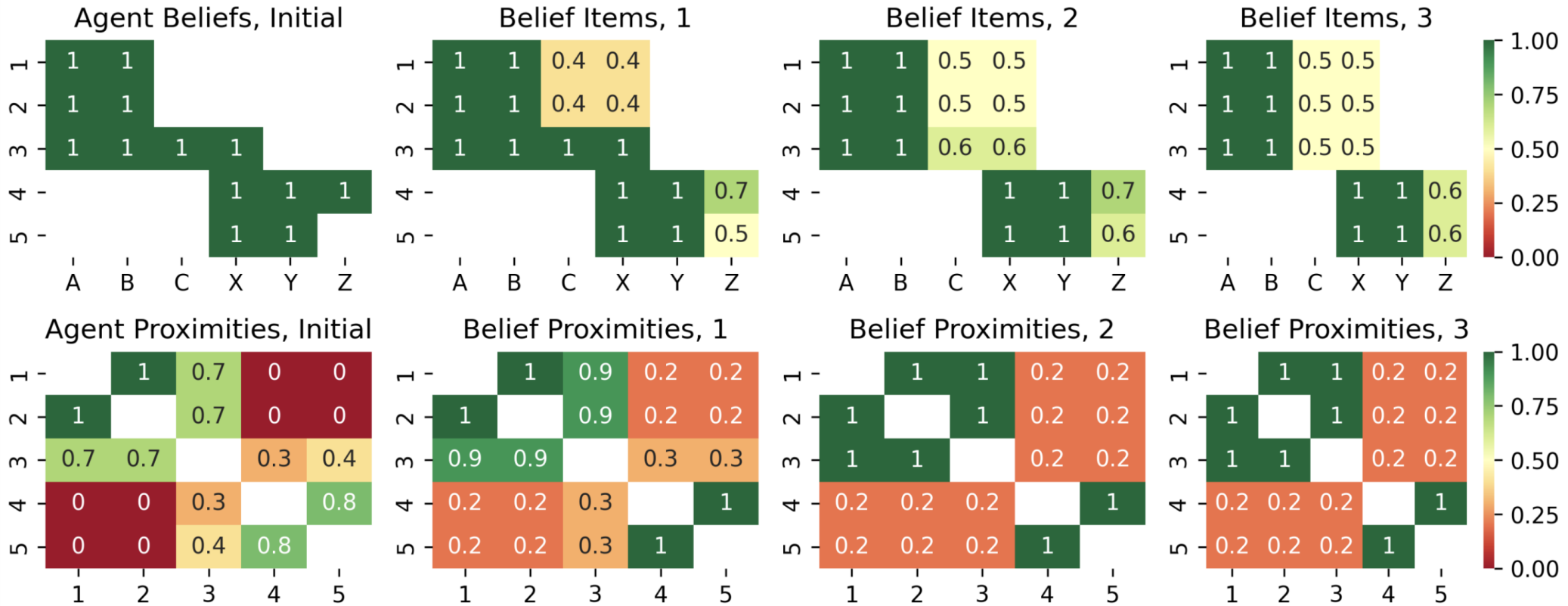
<https://ieeexplore.ieee.org/document/7361869>

<https://www.sciencedirect.com/science/article/pii/S1877050916317239>

https://link.springer.com/chapter/10.1007/978-3-319-97676-1_10

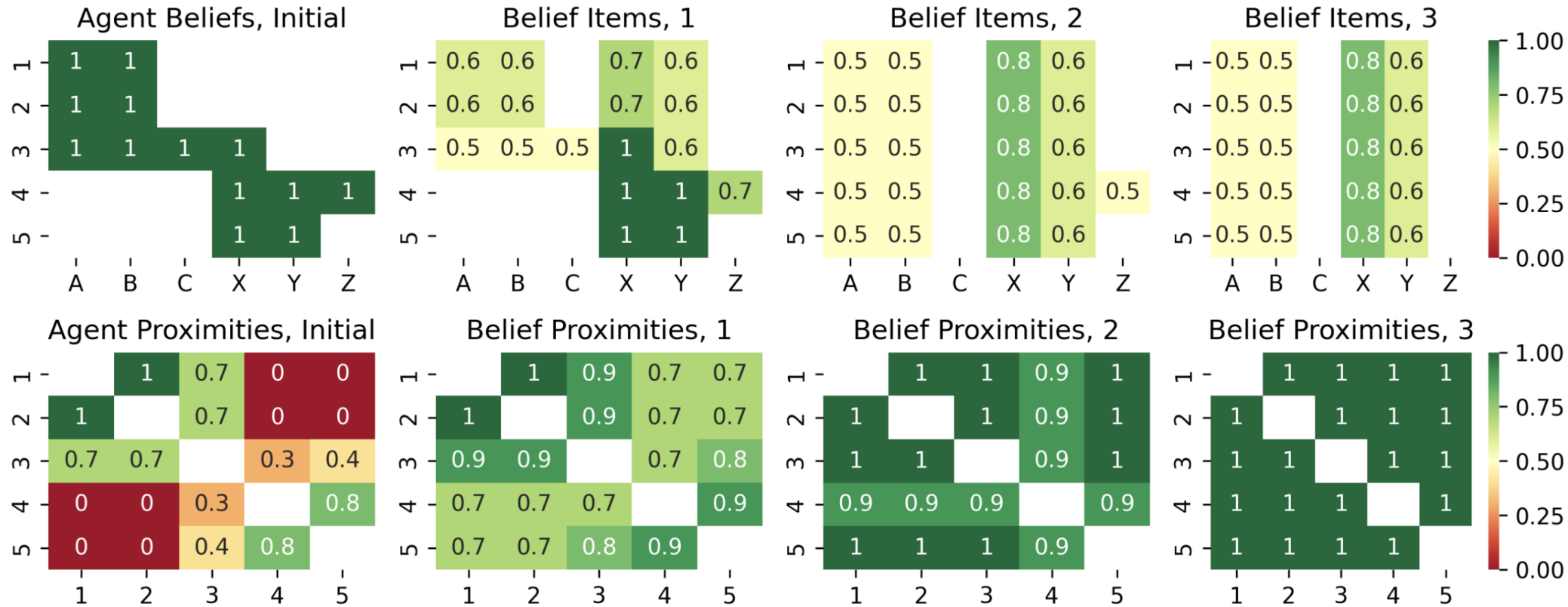
<https://link.springer.com/article/10.1186/s40708-026-00294-1> All rights reserved © 2026 Anton Kolonin, Aigents®

Simulation: socially limited, cognitively unlimited



Results of three rounds of multi-agent simulation with the forgetting threshold peer threshold = 0.5, the social relationship threshold forgetting threshold = 0.0 (limiting social connections K without constraint on belief capacity L). Top row: values of belief items A, B, C, X, Y, Z for five agents 1, 2, 3, 4, 5. Bottom row: social proximity matrices between agents, estimated as the cosine similarity of their beliefs. From left to right: the initial state before the simulation, and then the updated states of beliefs and proximities after three subsequent rounds of multi-agent communication.

Simulation: socially unlimited, cognitively limited



Results of three rounds of multi-agent simulation with the forgetting threshold forgetting threshold = 0.5, the social relationship threshold peer threshold = 0.0 (limiting belief capacity L without constraint on social connections K). Top row: values of belief items A, B, C, X, Y, Z for five agents 1, 2, 3, 4, 5. Bottom row: social proximity matrices between agents, estimated as the cosine similarity of their beliefs. From left to right: the initial state before the simulation, and then the updated states of beliefs and proximities after three subsequent rounds of multi-agent communication.

Consensus – technology to govern distributed multi-agent systems such as blockchains or societies, resistant to takeover and scam.

Proof-Of-Work



Force is Power:
Those who own more computing resources govern the network.

Proof-Of-Stake



Money is Power:
Those who have more money govern the network.

Proof-Of-Reputation



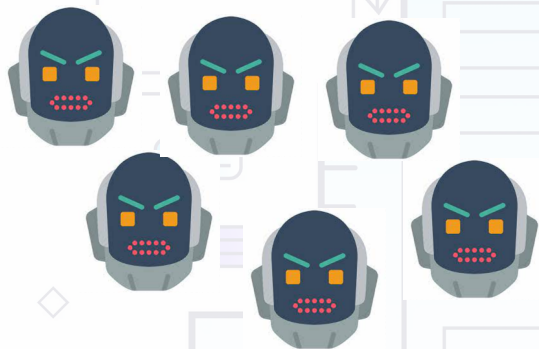
$$R_i = \sum_t \sum_j (R_j * V_{ijt})$$

Reputation is Power:
Those who earn a better reputation and a greater long-term audience base govern the network.

<https://steemit.com/blockchain/@agents/proof-of-reputation-as-liquid-democracy-for-blockchain>

Real Pain: Resisting Reputation Gaming

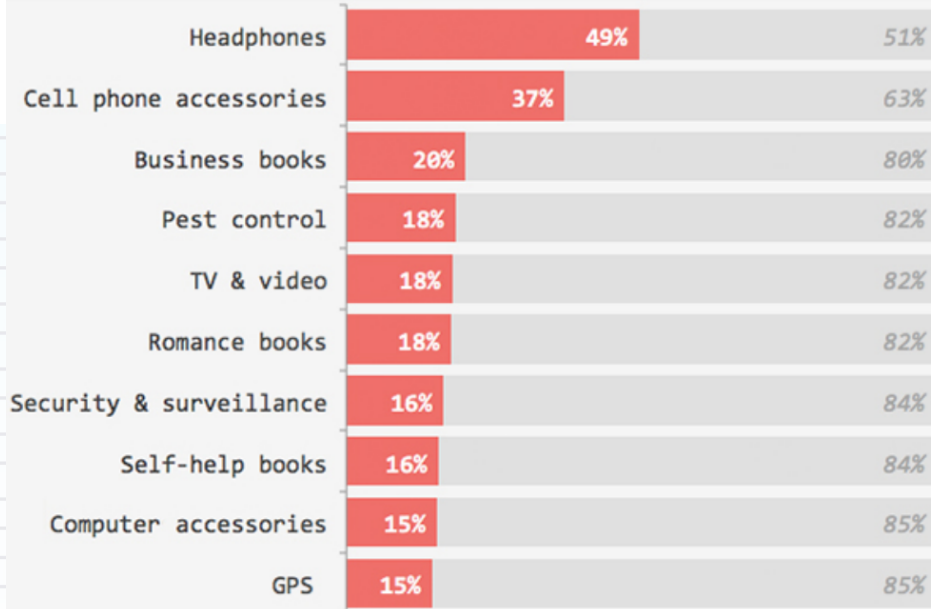
Cheaters



Fake accounts
with churning
and reputation
gaming bots

Amazon items with the highest % of fake reviews

Percent of total reviews identified as 'untrustworthy' by ReviewMeta's analysis tool (overall average = 11.3%)



Data via ReviewMeta; Limited to cats with 200k+ reviews
* Fake = simplified term for untrustworthy

the HUSTLE

Honest
vendor

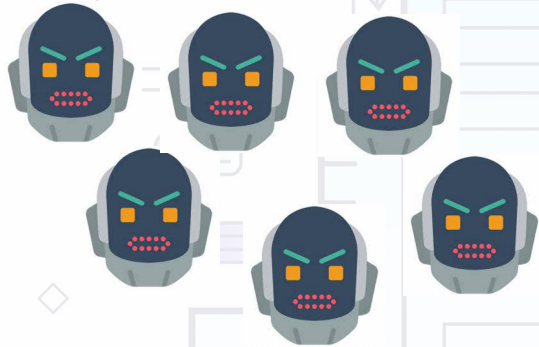


Honest
buyers

Bad reputation (rating) system



Cheaters
earn



Fake account
and bot farm
owners are
sponsors are
paid



Honest
vendors
lose profits

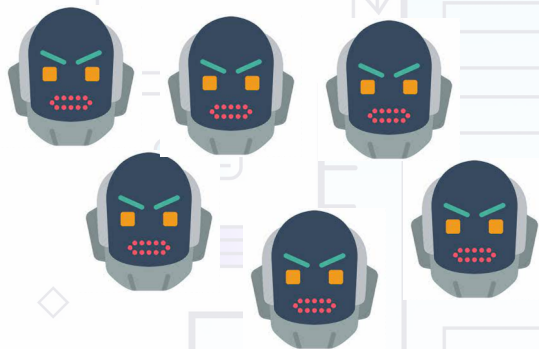


Honest
buyers lose
money

Right reputation (rating) system



Cheaters
lose
profits



Fake account and
bot farm owners are
sponsors are still
paid for no purpose

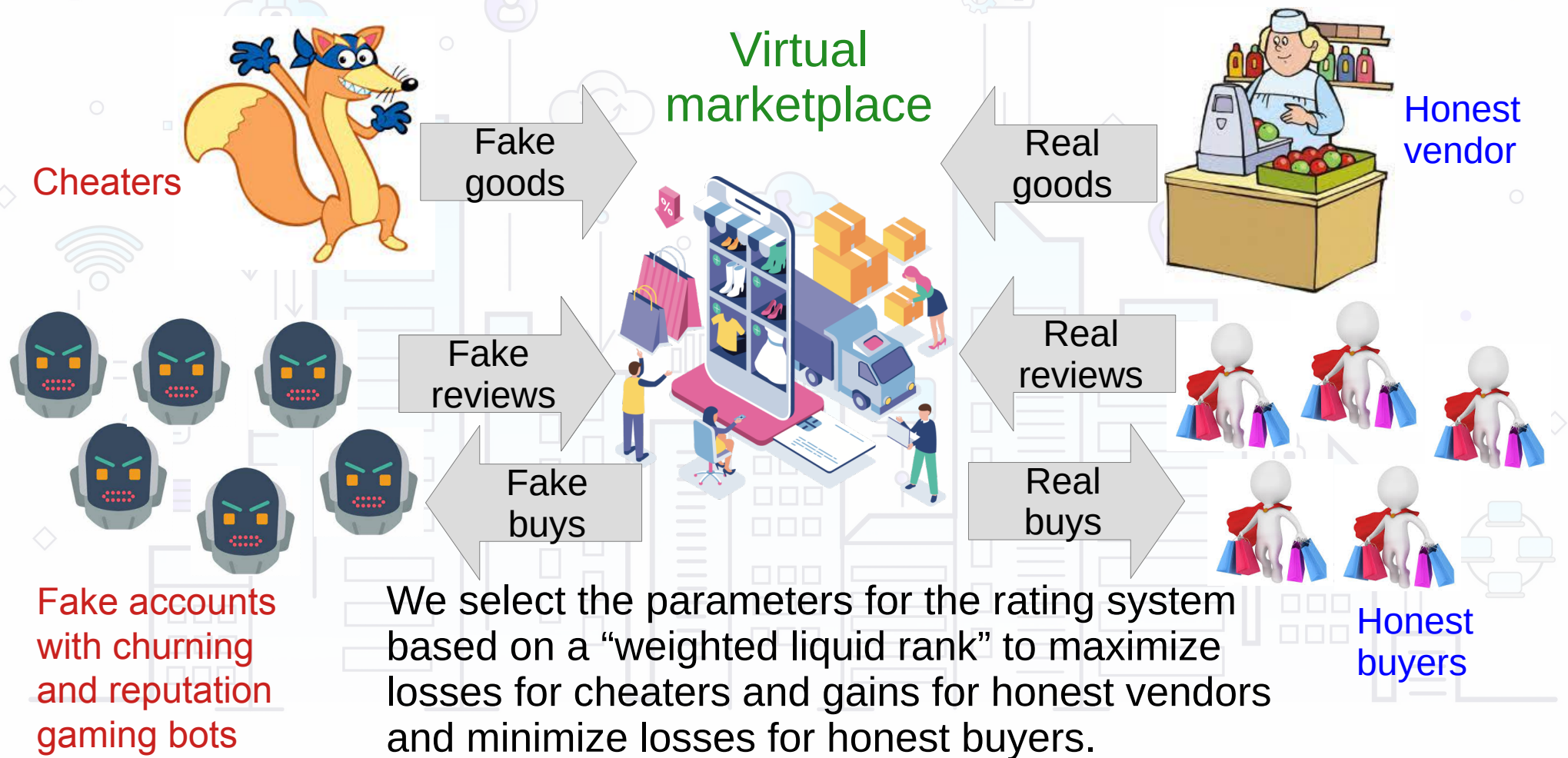


Honest
vendors
gain profits



Honest
buyers get
goods

Marketplace imitation modeling



**Algorithm 1** Weighted Liquid Rank (simplified version)**Inputs:**

- 1) Volume of rated transactions each with financial value of the purchased product or service and rating value evaluating quality of the product/service, covering specified period of time;
- 2) Reputation ranks for every participant at the end of the previous time period.

Parameters: List of parameters, affecting computations - default value, logarithmic ratings, conservatism, decayed value, etc.

Outputs: Reputation ranks for every participant at the end of the previous time period.

```

1: foreach of transactions do
2:   let rater_value be rank of the rater at the end of
     previous period of default value
3:   let rating_value be rating supplied by
     transaction rater (consumer) to ratee (supplier)
4:   let rating_weight be financial value of the
     transaction of its logarithm, if logarithmic ratings
     parameter is set to true
5:   sum rater_value*rating_value*rating_weight for
     every ratee
6: end foreach
  
```

```

7: do normalization of the sum of the multiplications
   per ratee to range 0.0-1.0, get differential_ranks
8: do blending of the old_ranks known at the end of
   previous period with differential_ranks based on
   parameter of conservatism, so that new_ranks =
   (old_ranks*conservatism+N*(1-differential_ranks)),
   using decayed value if no rating are given to ratee
   during the period
9: do normalization of new_ranks to range 0.0-1.0
10:return new_ranks
  
```

- R_d - default initial reputation rank;
- R_e - decayed reputation in range to be approached by inactive agents eventually;
- C - conservatism as a blending "alpha" factor between the previous reputation rank recorded at the beginning of the observed period and the differential one obtained during the observation period;
- *FullNorm* – when this boolean option is set to *True* the reputation system performs a full-scale normalization of incremental ratings;
- *LogRatings* - when this boolean option is set to *True* the reputation system applies $\log_{10}(1+value)$ to financial values used for weighting explicit ratings;
- *Aggregation* - when this boolean option is set to *True* the reputation system aggregates all explicit ratings between each unique combination of two agents with computes a weighted average of ratings across the observation period;
- *Downrating* - when this boolean option is set to *True* the reputation system translates original explicit rating values in range 0.0-0.25 to negative values in range -1.0 to 0.0 and original values in range 0.25-1.0 to the interval 0.0-1.0.
- *UpdatePeriod* – the number of days to update reputation state, considered as observation period for computing incremental reputations.



SingularityNET

<https://singularitynet.io>

Reputation System for Marketplaces

<https://medium.com/singularitynet/minimizing-recommendation-fraud-7dabbee8fc00>

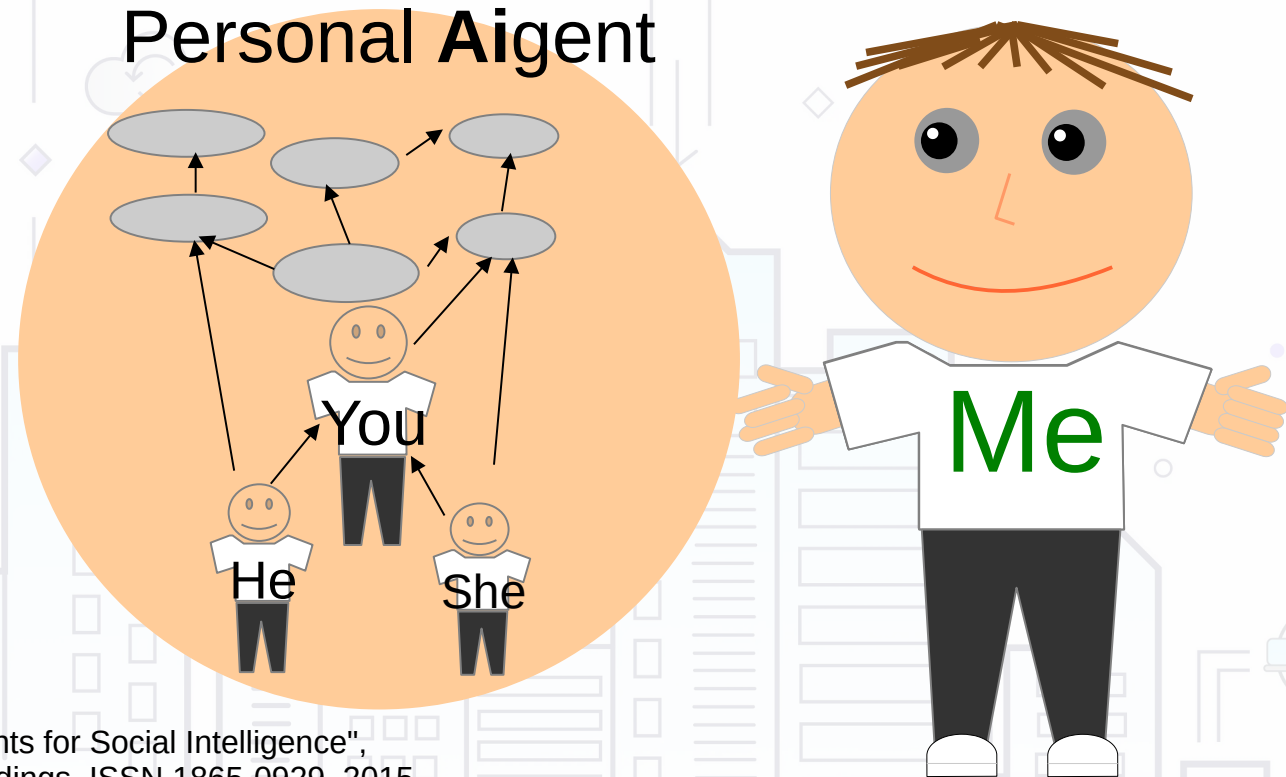
Using Reputation System for protection from scam identifying dishonest suppliers.



Ranks of Suppliers, dishonest Supplier (including alias) in red and honest suppliers in blue

Aigents® - Personal Assistant

Personal Aigent



A. Kolonin, "Aigents: Adaptive Personal Agents for Social Intelligence",
KESW, 6th International Conference Proceedings, ISSN 1865-0929, 2015
https://link.springer.com/chapter/10.1007/978-3-319-24543-0_22

A.Kolonin, "Adaptive experiential learning for business intelligence
agents", CSGB Symposium Proceedings, 2016
<https://ieeexplore.ieee.org/document/7587679>

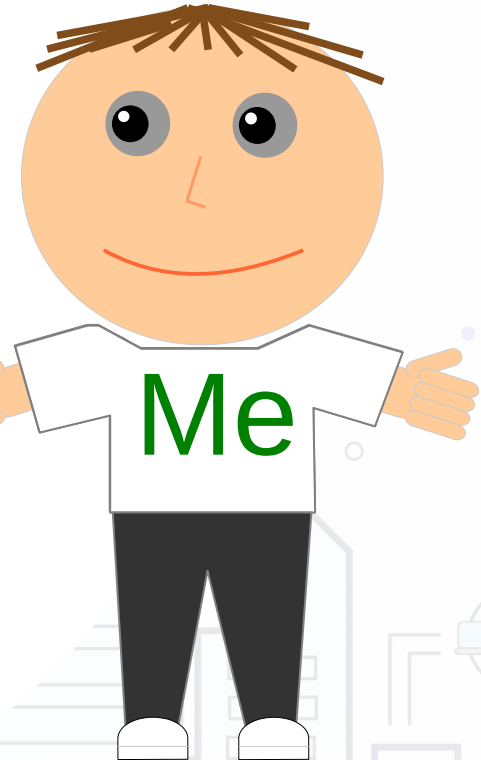
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2015-2016

Aigents® - Social computing platform



News



Ethereum



PayPal



Steemit



Golos.id



Google



Vkontakte



Facebook



Messenger



Slack



Telegram



Reddit



Friends

Twitter



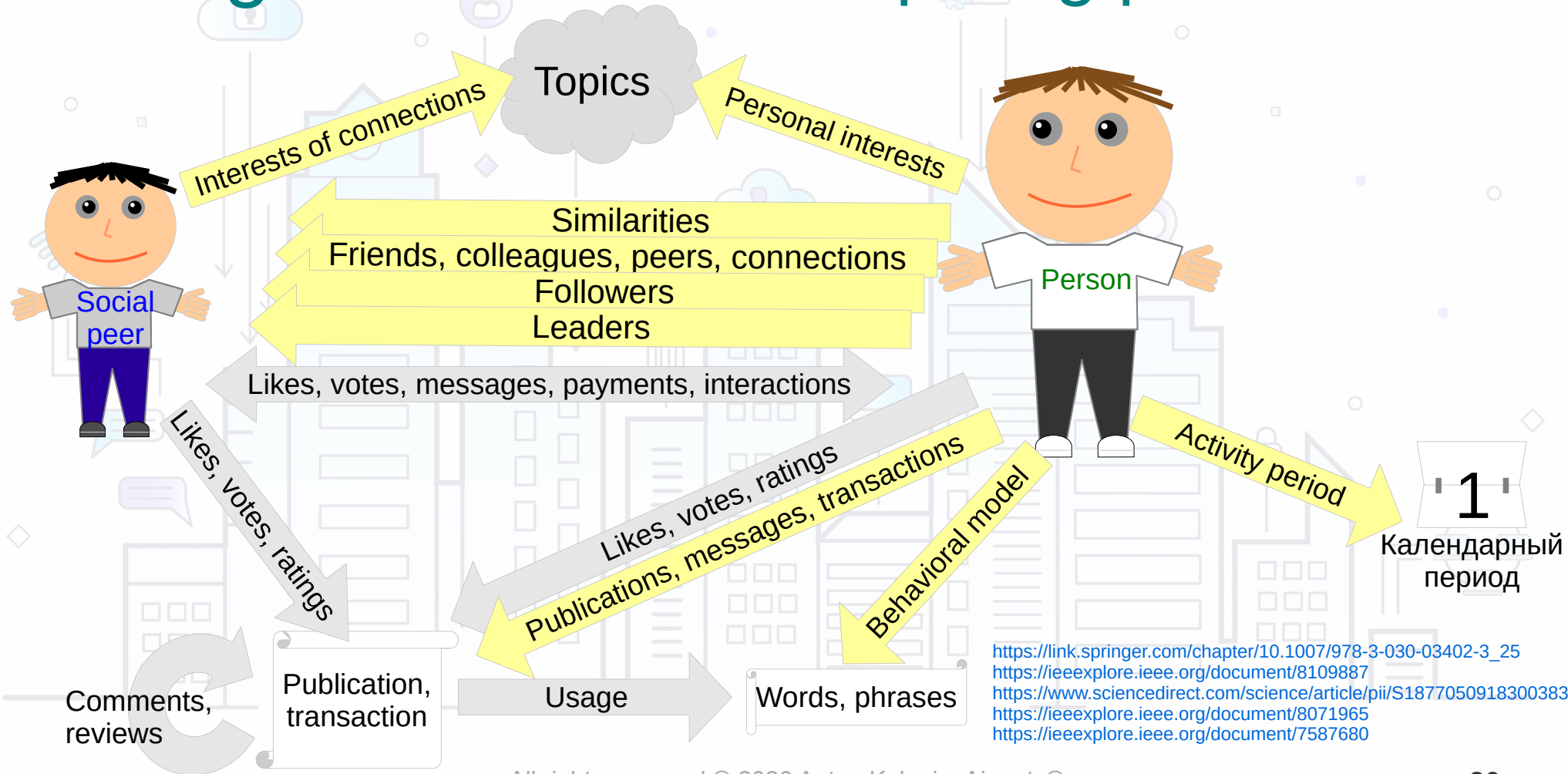
RSS



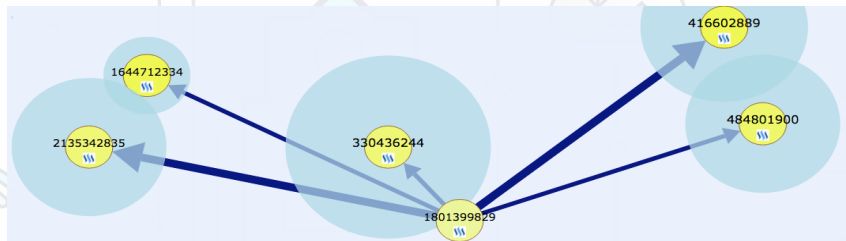
Discourse



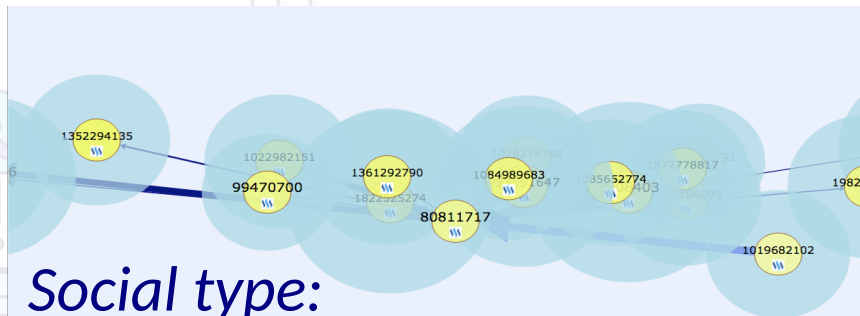
Aigents® - Social computing platform



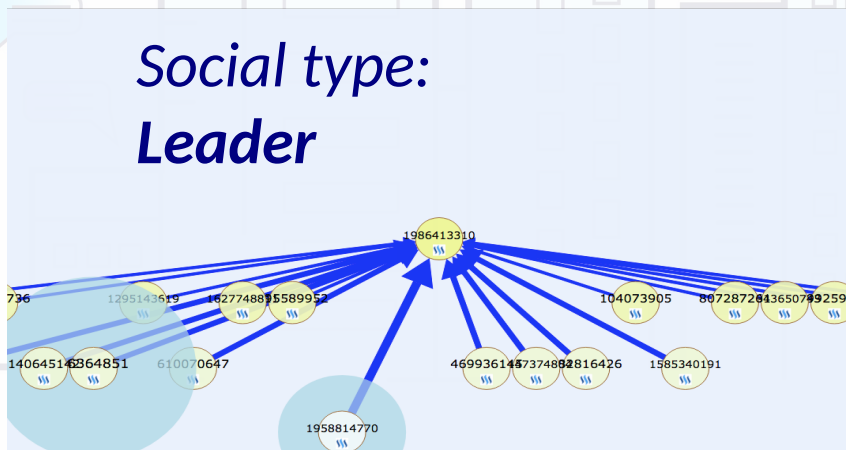
Solution: Identification and discovery of socio-communicative types, profiles and structures, social circles and audiences, opinion leaders, and information dissemination channels.



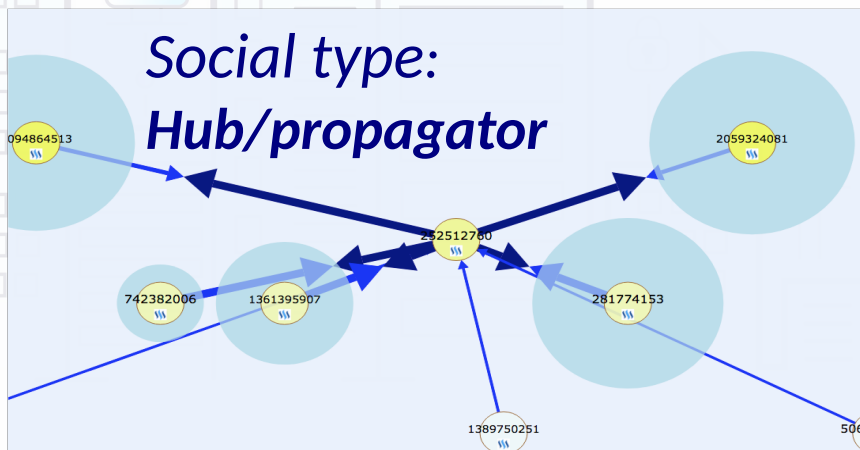
Social type:
Follower



Social type:
Peer/colleague/friend

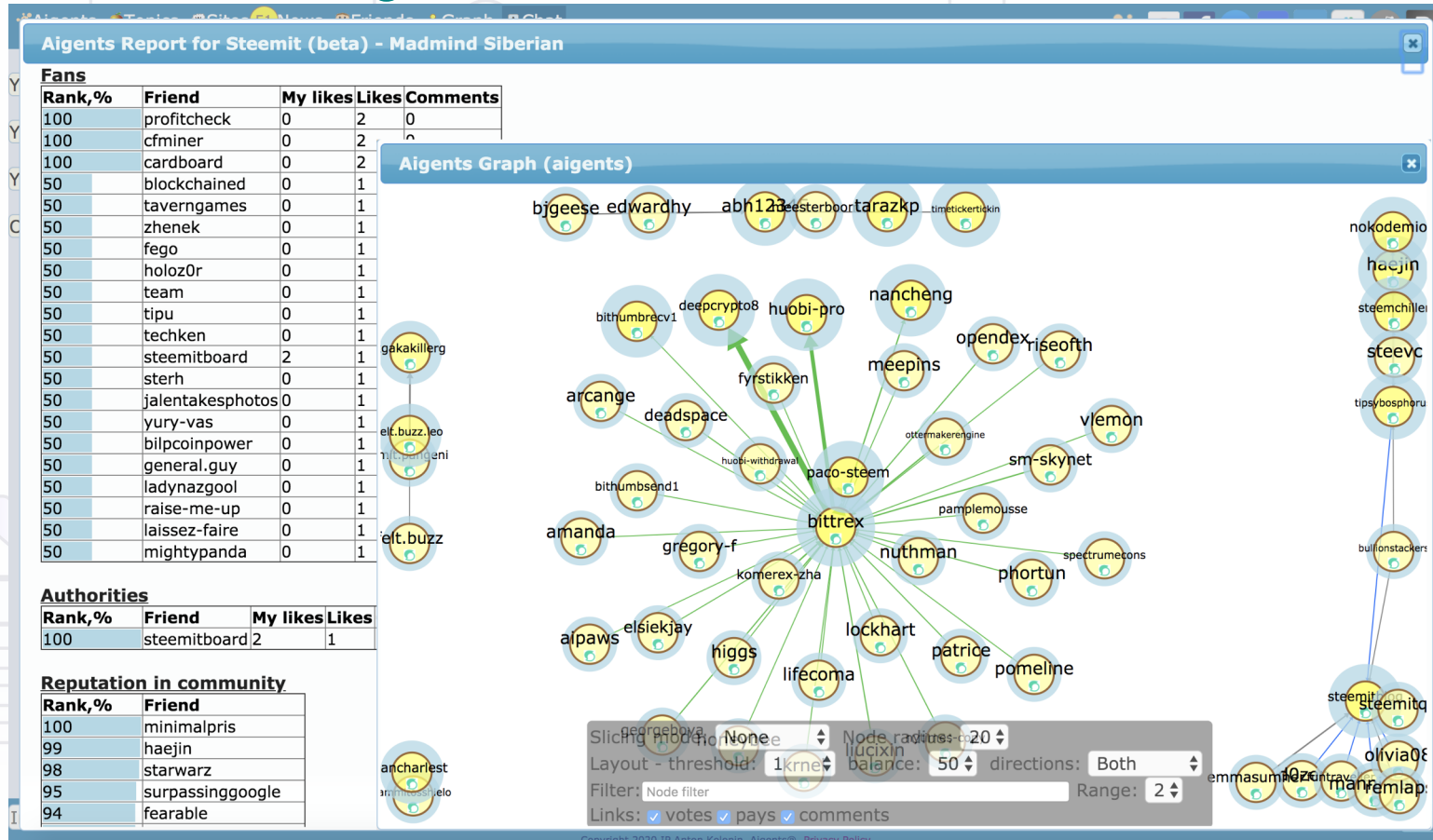


Social type:
Leader



Social type:
Hub/propagator

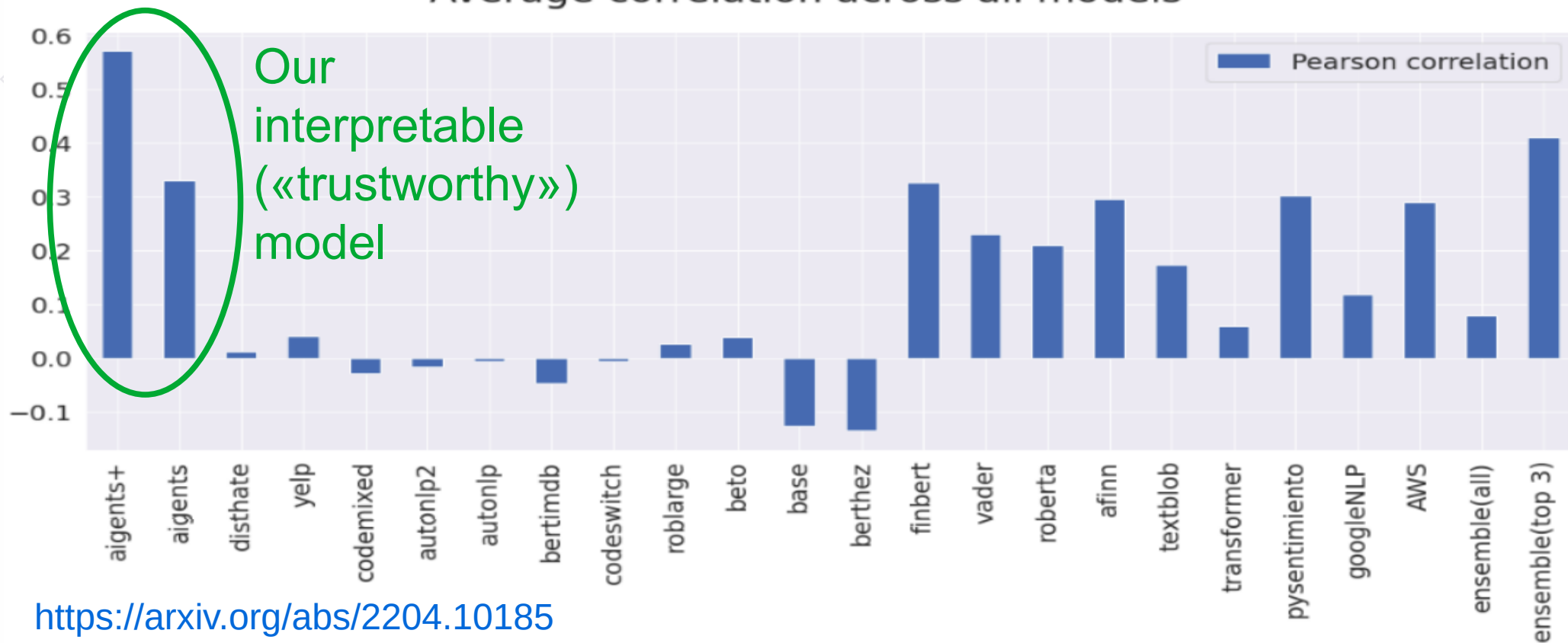
Solution: Quantitative assessment of reputation and trust levels, as well as the nature, quantity, and quality of interpersonal relationships within large-scale online communities.



How do you determine the tone or sentiment of a text?

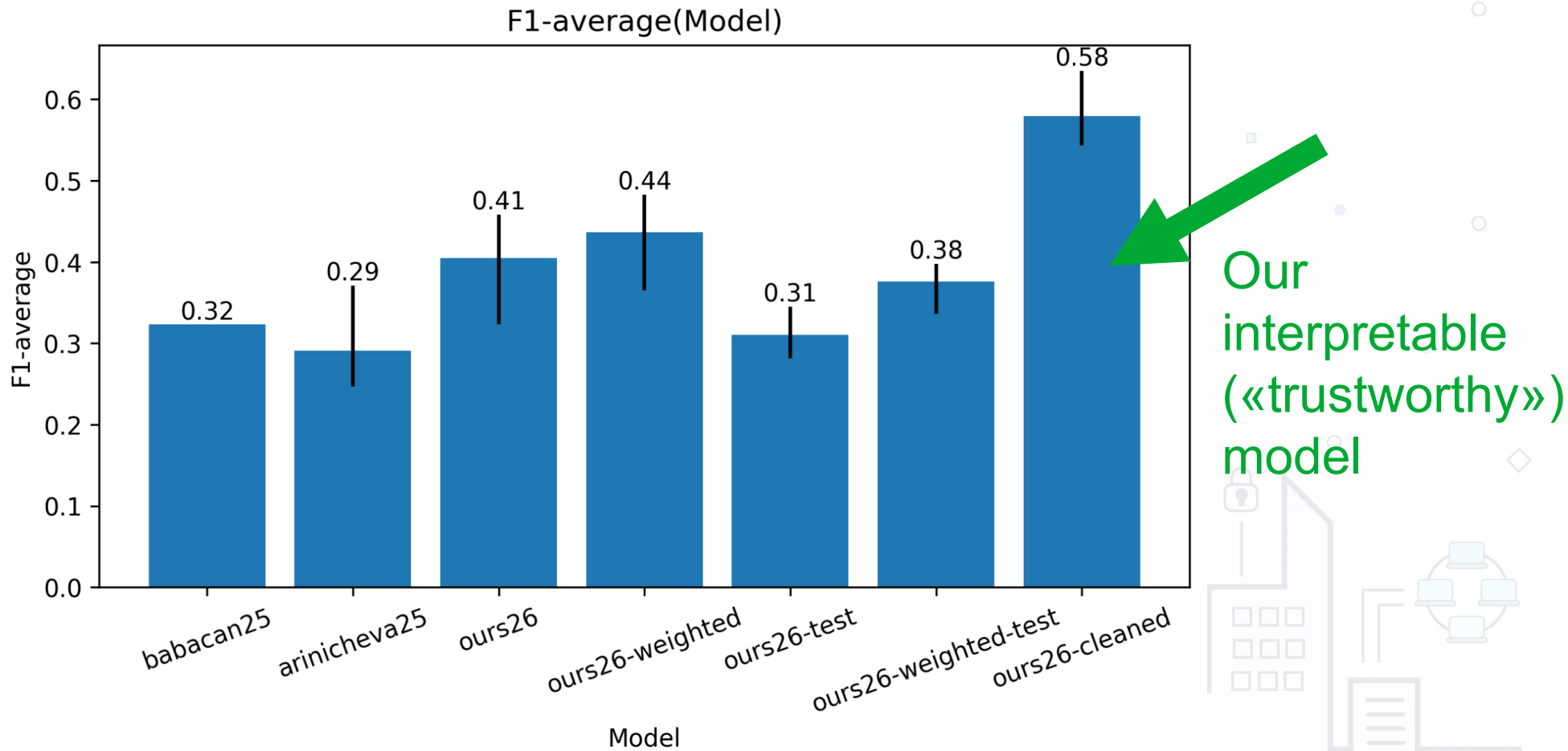
Metrics: Sentiment, Positiveness, Negativeness, Contradictiveness

Average correlation across all models



<https://arxiv.org/abs/2204.10185>

Automated detection of cognitive distortions - state of the art

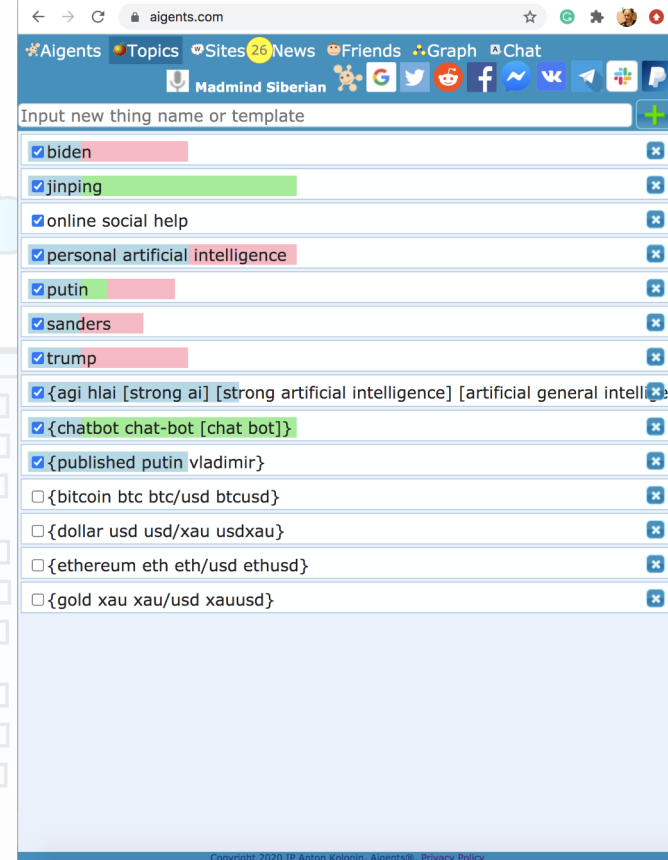
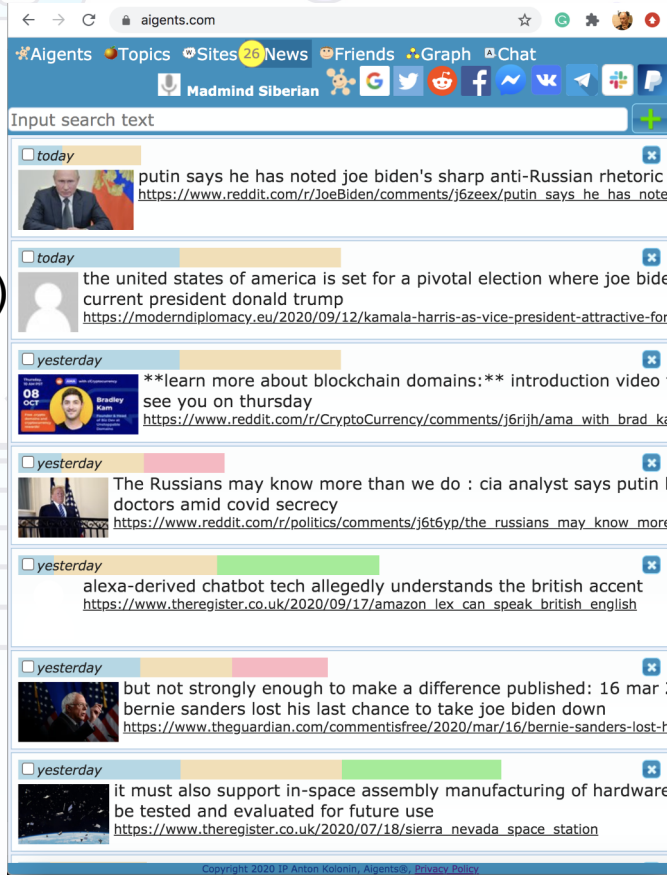


<https://peerj.com/articles/cs-3699/>

<https://arxiv.org/abs/2511.05969>

Solution: News filtering, recommendation, and tagging based on sentiment, cognitive distortions and social relevance

News with positive (green) and negative (red) sentiment and social relevance (yellow)



Topics by news with positive (green) and negative (red) sentiment

<https://blog.singularitynet.io/aigents-sentiment-detection-personal-and-social-relevant-news-be989d73b381>

Automated detection of cognitive distortions, emotions and social situations

P: Hi! How are you?

K: Bad. They have kicked out the loser! (tentatively diagnosing #being fired and #personalization)

P: How did that happen? Who did that? (clarifying social situation)

K: It was my job. I am fired so to speak. (confirming #being fired)

P: Does it make you anxious or make you angry on them? (clarifying emotion - #anxiety or #anger)

K: Anxious, of course. I have to pay the mortgage. (confirming #anxiety, diagnosing #shouldment)

P: You have been fired from your job and you are afraid, that won't be able to pay, right? (#being fired, #anxiety)

K: Yes. (confirming diagnosis in respect to social situation and emotion)

P: What do you think was the reason?

K: I just was not lucky – the investor has left the company. (diagnosing #personalization)

P: Was that your fault? (clarifying cognitive distortion - #personalization)

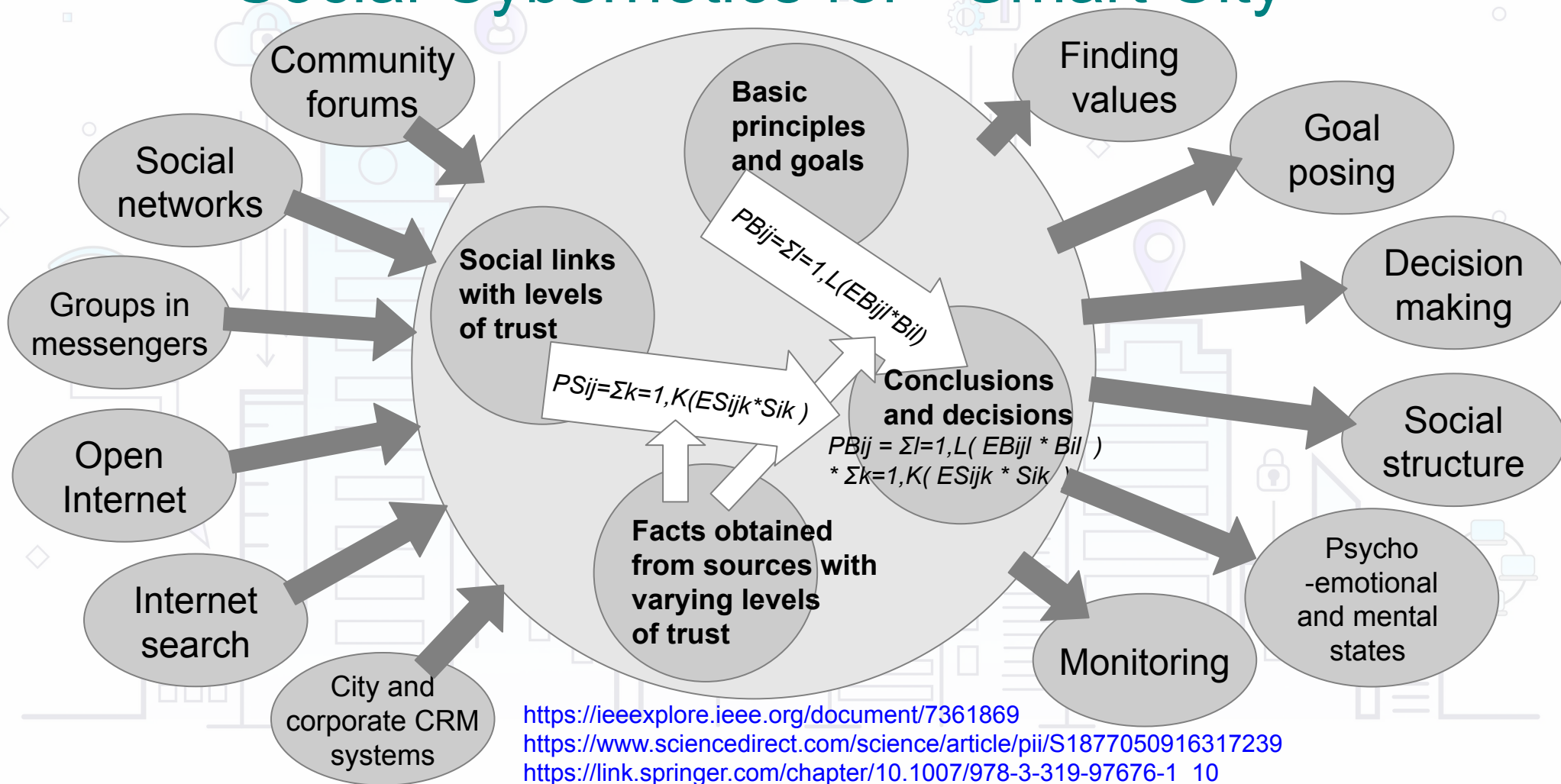
K: Not at all. I just feel like my life is ruined. (change diagnosis to #catastrophizing)

P: You are anxious, that after the being fired you won't be able to find a new job, correct? (#anxiety, #being fired, #catastrophizing)

K: Exactly. Although in fact I am an experienced professional. (confirming diagnosis in regard to emotion, social situation and cognitive distortion moving on to correction with healing interventions)

<https://doi.org/10.1016/j.procs.2022.11.054>

Social Cybernetics for «Smart City»



<https://ieeexplore.ieee.org/document/7361869>

<https://www.sciencedirect.com/science/article/pii/S1877050916317239>

https://link.springer.com/chapter/10.1007/978-3-319-97676-1_10

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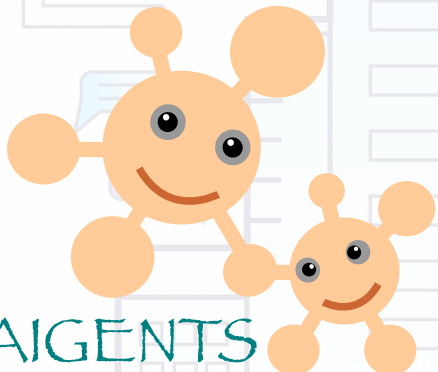
Thank you for attention!

More on YouTube:

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Telegram: akolonin



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University
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